

Pericyclic reactions 2: sigmatropic and electrocyclic reactions

35

Connections

➡ Building on

- Cycloadditions and the principles of pericyclic reactions (essential reading!) **ch34**
- Acetal formation **ch11**
- Conformational analysis **ch16**
- Elimination reactions **ch17**
- Controlling alkene geometry and main group chemistry **ch27**
- The synthesis of aromatic heterocycles **ch30**

Arriving at

- The second and third types of pericyclic reaction
- Stereochemistry from chair-like transition states
- What decides whether these pericyclic reactions go 'forwards' or 'backwards'
- Special chemistry of N, S, and P
- Why substituted cyclopentadienes are unstable
- What 'con'- and 'dis'-rotatory means

➡ Looking forward to

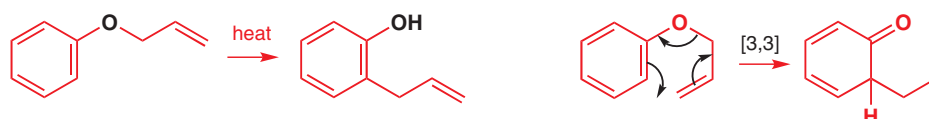
- Rearrangements **ch36**
- Asymmetric synthesis **ch41**
- Natural products **ch42**

Cycloadditions, the subject of the last chapter, are just one of the three main classes of pericyclic reaction. In this chapter we consider the other two classes: **sigmatropic rearrangements** and **electrocyclic reactions**. We will analyse them in a way that is similar to our dealings with cycloadditions.

Sigmatropic rearrangements

The Claisen rearrangement was the first to be discovered

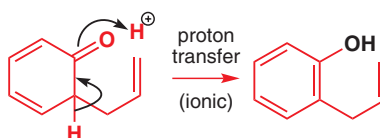
The original sigmatropic rearrangement occurred when an aryl allyl ether was heated without solvent and an *ortho*-allyl phenol resulted. This is the Claisen rearrangement. The first step in this reaction is a pericyclic reaction of a type that you will learn to call a [3,3]-sigmatropic rearrangement.



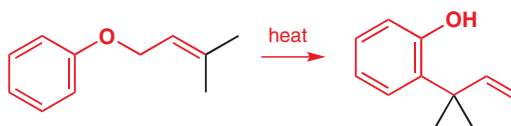
 Interactive mechanism for aromatic Claisen rearrangement

This is a one-step mechanism without ionic intermediates or any charges, just like a cycloaddition. The arrows go round in a ring. The difference between this and a cycloaddition is that one of the arrows starts on a σ bond instead of on a π bond. The second step in the reaction is a simple ionic proton transfer to regenerate aromaticity.

Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.



How do we know that this is the mechanism? If the allyl ether is unsymmetrical, it turns 'inside out' during Claisen rearrangement, as required by the mechanism. Check for yourself that this is right.

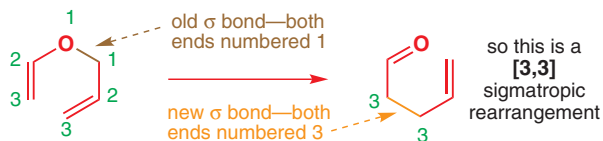


The aliphatic Claisen rearrangement also occurs

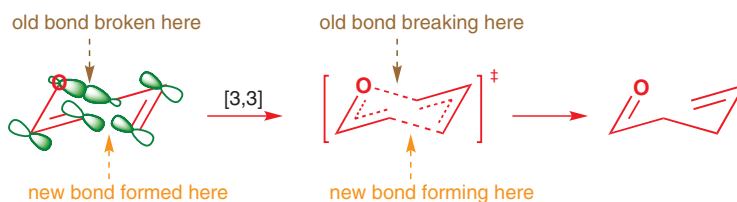


It was later found that the same sort of reaction occurs without the aromatic ring. This is called either the aliphatic Claisen rearrangement or the Claisen–Cope rearrangement. Here is the simplest possible example.

All these reactions are called sigmatropic because a σ bond appears to move from one place to another during the reaction. This particular reaction is called a [3,3]-sigmatropic rearrangement because the new σ bond has a 3,3 relationship to the old σ bond. You can see this if you number both ends of the old σ bond '1' and count round in both directions to the ends of the new σ bond in the product. You will find that the ends of the new σ bond both have the number '3'.



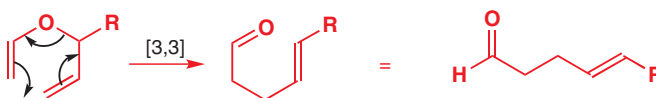
These [3,3]-sigmatropic rearrangements happen through a chair-like transition state, which allows us both to get the orbitals right and to predict the stereochemistry (if any) of the new double bond. The orbitals look something like this.



Note that these do not represent any specific frontier orbitals, they simply show that, in this conformation, the new σ bond is formed from two p orbitals that point directly at each other and that the two new σ bonds are formed from orbitals that are already parallel.

Alkene stereochemistry in the Claisen rearrangement comes from a chair-like transition state

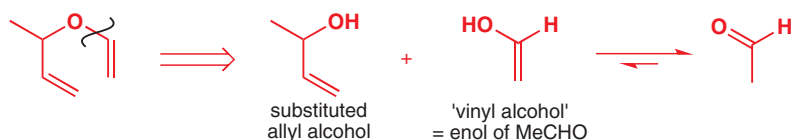
Stereochemistry may arise if there is a substituent on the saturated carbon atom next to the oxygen atom. If there is, the resulting double bond strongly favours the *trans* (*E*) geometry. This is because the substituent prefers an equatorial position on the chair transition state.



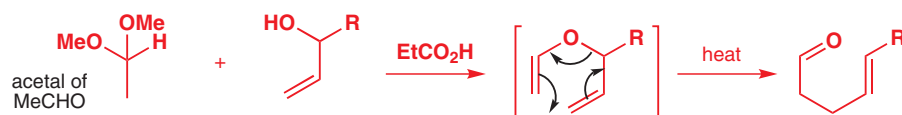
The substituent R prefers an equatorial position as the molecule reacts and R retains this position in the product. The new alkene bond is shown in green. Notice that the *trans* geometry of the alkene in the product is already there in the conformation chosen by the starting material and in the transition state.



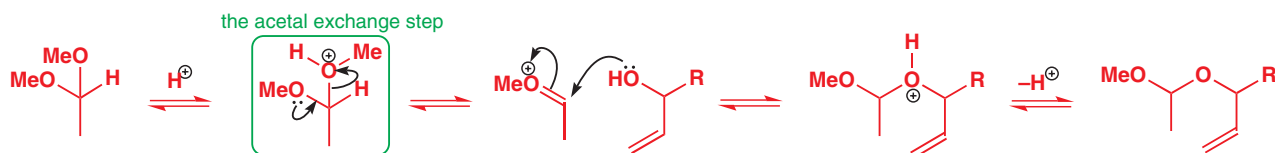
The starting material for these aliphatic Claisen rearrangements consists of ethers with one allyl and one vinyl group. We need now to consider how such useful molecules might be made. There is no problem about the allyl half—allylic alcohols are stable, easily made compounds. But what about the vinyl half? 'Vinyl alcohol' is just the enol of acetaldehyde (MeCHO).



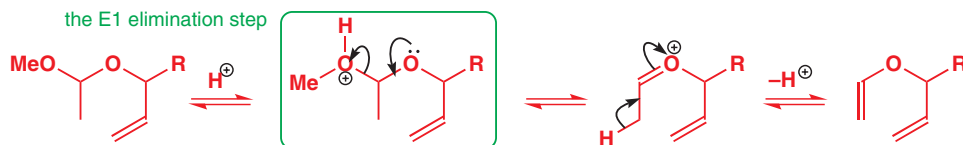
The solution is to use an acetal of the aldehyde in an acid-catalysed exchange process with the allylic alcohol. It is not necessary to isolate the allyl vinyl ether as long as some of it is formed and rearranges into the final product.



The acid catalyst usually used, propanoic acid, has a conveniently high boiling point so that the whole mixture can be equilibrated at high temperature. The first step is an acetal exchange in which the allylic alcohol displaces methanol.



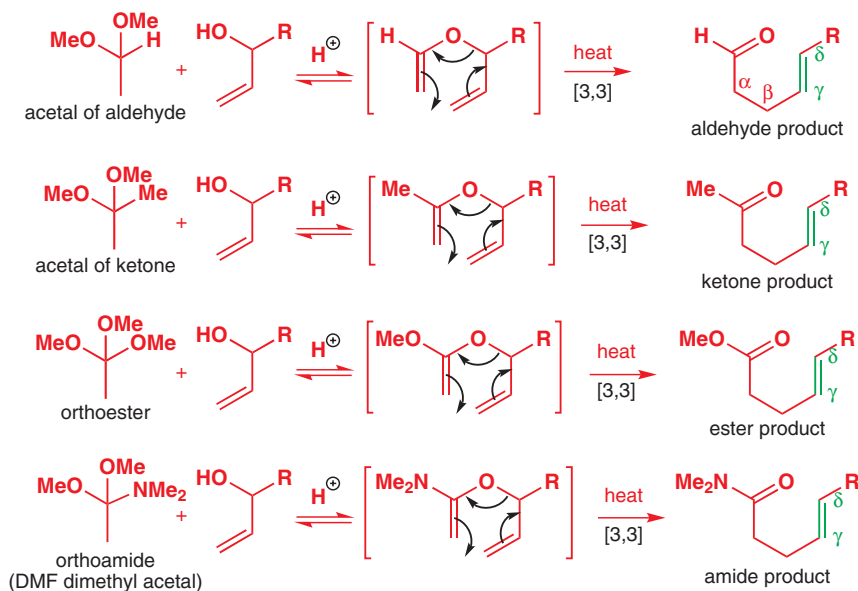
The methanol is distilled off as it is the most volatile of the components in this mixture. A second molecule of methanol is now lost in an acid-catalysed elimination reaction to give the vinyl group.



■ Note that the first molecule of methanol was displaced in an S_N1 reaction and the second lost in an E1 reaction. The chemistry of acetals is dominated by the loss of protonated OR or OH groups in the steps with green boxes. Never be tempted to write S_N2 mechanisms with acetals.

The Claisen rearrangement is a general synthesis of γ,δ -unsaturated carbonyl compounds

The [3,3]-sigmatropic rearrangement itself can be carried out by heat as part of the same step or as a separate step depending on the compounds. This is a very flexible reaction sequence and can be used for aldehydes (as shown above), ketones, esters, or amides. In each case acetal-like compounds are used—acetals themselves for aldehydes and ketones; orthoesters and orthoamides for the other two (although the orthoamides are often called 'amide acetals').



The common feature in the products of these Claisen rearrangements is a γ,δ -unsaturated carbonyl group. If this is what you need in a synthesis, make it by a Claisen rearrangement.

Orbital descriptions of [3,3]-sigmatropic rearrangements

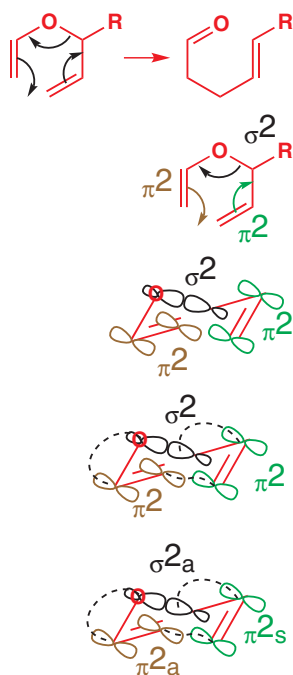
➡ If you need reminding of the meanings of the terms or symbols in this section please turn back to p. 892 of Chapter 34 now.

It is possible to give a frontier orbital description of a [3,3]-sigmatropic rearrangement but this is not a very satisfactory treatment because we don't have two separate reagents recognizing each other across space as we did in cycloadditions. There are *three* components in these reactions—two non-conjugated π bonds that do have to overlap across space and a σ bond in the chain joining the two π bonds. The Woodward–Hoffmann rules give a more satisfying description and we shall follow the routine outlined on p. 892 for cycloadditions. Note that for stage 3, we can use the three-dimensional diagram we have already made.

First a reminder of the Woodward–Hoffmann rules:

• The Woodward–Hoffmann rules

In a thermal pericyclic reaction the total number of $(4q+2)_s$ and $(4r)_a$ components must be odd.



1. Draw the mechanism for the reaction (we shall stay with a familiar one).
2. Choose the components. All the bonds taking part in the mechanism must be included and no others.
3. Make a three-dimensional drawing of the way the components come together for the reaction, putting in orbitals at the ends of the components (only!). Note that we have dropped the shading in the orbital from the previous diagrams earlier in the chapter.
4. Join up the components where new bonds are to be formed. Make sure you join orbitals that are going to form new bonds.
5. Label each component s or a depending whether new bonds are formed on the same or on opposite sides. See below for the σ bond symmetry.

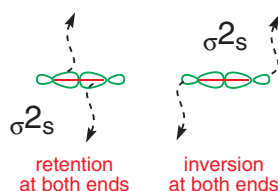
6. Add up the number of $(4q+2)_s$ and $(4r)_a$ components. If the sum is odd, the reaction is allowed. Here there is:
 one $(4q+2)_s$ component (one alkene) and
 no $(4r)_a$ components.

Total=1, so this is an allowed reaction. As you saw in Chapter 34 (p. 893), the $\pi 2_a$ and $\sigma 2_a$ components have irrelevant symmetry and are not counted.

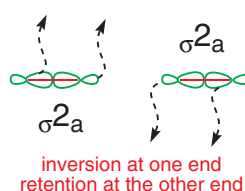
One new aspect of orbital symmetry has appeared in this diagram—how did we deduce a *s* or *a* symmetry in the way the σ bond reacted? For π bonds it is simple—if both bonds are formed on the same side of the old π bond, it has reacted suprafacially; if on opposite sides, antarafacially.

With a σ bond the symmetry is not so obvious. We want to know if it does the *same* thing at each end (*s*) or a *different* thing (*a*). But what is the ‘thing’ it does? It reacts using the large lobe of the sp^3 orbital (retention) or the small lobe (inversion). If it reacts with retention at both ends or inversion at both ends, it reacts suprafacially, while if it reacts with retention at one end and inversion at the other, it reacts antarafacially. There are four possibilities.

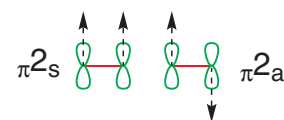
σ bond reacting suprafacially



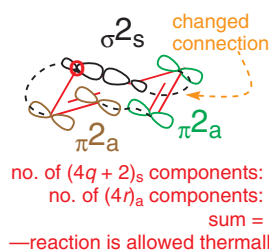
σ bond reacting antarafacially



In the routine above, we chose to use our σ bond so that we got inversion at one end and retention at the other. That was why we identified it as an antarafacial component. If we had chosen another style we should have got different descriptions of the components, but the reaction would still have been allowed, for example changing just one connecting line, as in the margin, changes the symmetry of the σ bond so that it becomes a $\sigma 2_s$ component but it also changes the symmetry of one of the π bonds so that it becomes a $\pi 2_a$ component. The net result is still only one component of the Woodward–Hoffmann symmetry, the sum is still 1, and the reaction still allowed.



■ If you are interested in the frontier orbital approach to [3,3]-sigmatropic reactions, you could read about it in Ian Fleming (2009) *Frontier orbitals and organic chemical reactions*, 2nd edn, Wiley, Chichester. We shall use this approach when we come to [1,5]-sigmatropic rearrangements.



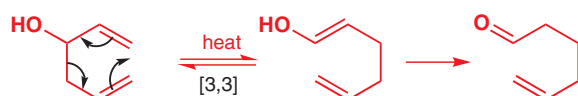
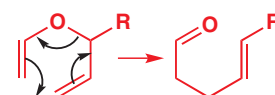
The direction of [3,3]-sigmatropic rearrangements

Orbital symmetry tells us that [3,3]-sigmatropic rearrangements are allowed but says nothing about which way they will go. They are allowed in either direction. So why does the Claisen–Cope rearrangement always form the carbonyl-containing product? Think back to our discussion on enols (Chapter 20) and you may recall that the combination of a carbonyl group and a C–C σ bond made the keto form more stable than the enol form with its combination of a C=C π bond and a C–O σ bond. The same is true here. It is the stability of the carbonyl group that drives the reaction to the right.

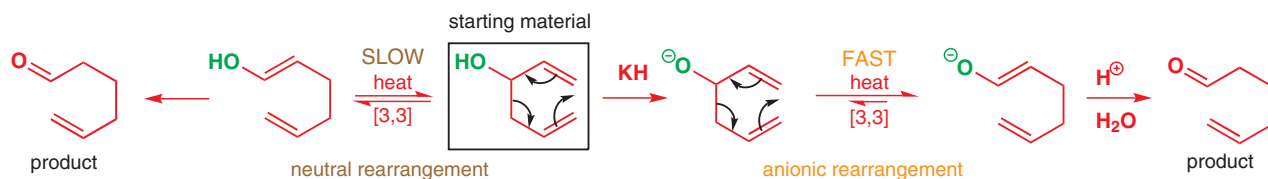
Directing the Cope rearrangement by the formation of a carbonyl group

The Cope rearrangement is a [3,3]-sigmatropic rearrangement with only carbon atoms in the ring. In its simplest version it is not a reaction at all. The starting material and the product are the same.

We can drive this reaction too by the formation of a carbonyl group if we put an OH substituent in the right place.

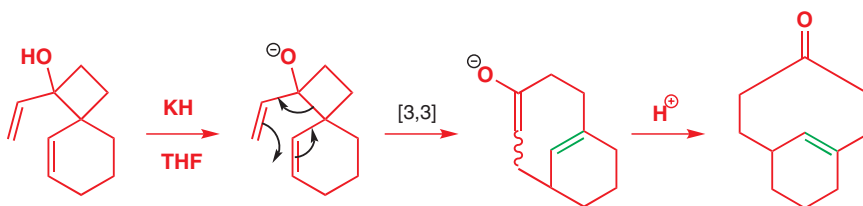


The product of the sigmatropic step is the enol of the final product. It turns out that the reaction is accelerated if the starting alcohol is treated with base (KH is the best) to make the alkoxide. The product is then the potassium enolate, which is more stable than the simple potassium alkoxide starting material. As the reaction proceeds, conjugation is growing between O^- and the new π bond.



► Bredt's rule forbidding bridgehead alkenes and the reasons for it are discussed in Chapter 17.

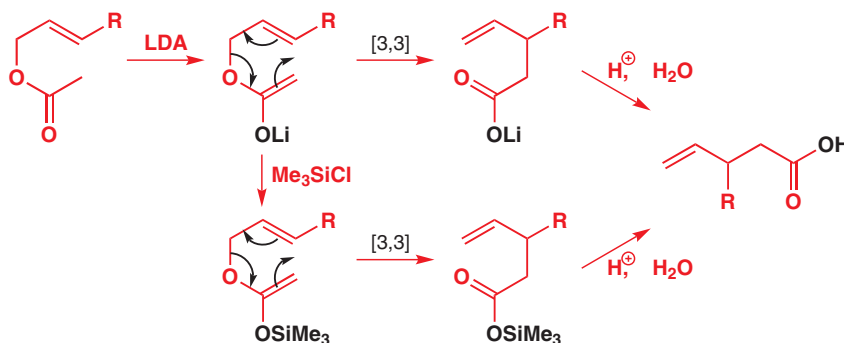
Some remarkable compounds can be made by this method. One of the strangest—a 'bridge-head' alkene—was made by a potassium alkoxide-accelerated Cope rearrangement in which a four-membered ring was expanded into an eight-membered ring containing a *trans* double bond (shown in green).



A combination of an oxygen atom in the ring and another one outside the ring is very powerful at promoting [3,3]-sigmatropic rearrangements and easy to arrange by making the lithium enolate of an ester of an allylic alcohol.

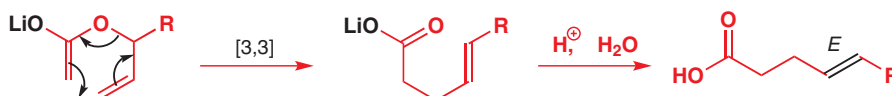


Sometimes it is better to convert the lithium enolate into the silyl enol ether before heating to accomplish the [3,3]-sigmatropic rearrangement. In any case, both products give the unsaturated carboxylic acid on work-up.



Interactive mechanism for Ireland-Claisen rearrangement

This reaction is known as the Ireland-Claisen rearrangement as it was a variation of the Claisen rearrangement invented by R. E. Ireland in the 1970s and widely used since. If the substituents are suitably arranged, it shows the same *E* selectivity as the simple Claisen rearrangement and for the same reason.

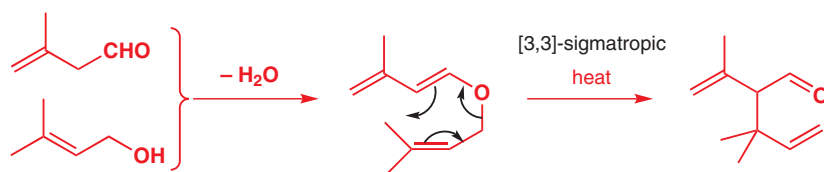
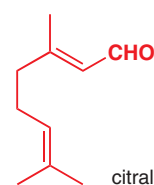


In some cases simple Cope rearrangements without any oxygen atoms at all can be directed by an unstable starting material or a stable product. The instability might be strain and the stability might simply be more substituents on the double bonds. In the next reaction the driving force is the breaking of a weak σ bond in a three-membered ring. This reaction goes in 100% yield at only just above room temperature, so it is very favourable. In the second example, the trisubstituted double bonds inside the five-membered rings of the product are more stable than the exomethylene groups in the starting material.



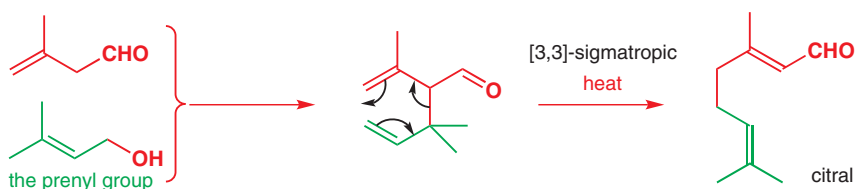
An industrial synthesis of citral

'Citral' is a key intermediate in the synthesis of vitamin A, and it is manufactured by a remarkable process that involves two successive [3,3]-sigmatropic rearrangements, a Claisen followed by a Cope. The allyl vinyl ether needed for the Claisen rearrangement is an enol ether of an unsaturated aldehyde with an unsaturated alcohol. The two starting materials are themselves derived from a common precursor, making this a most efficient process! Heating the enol ether promotes [3,3]-sigmatropic rearrangement propelled by the formation of a carbonyl group.



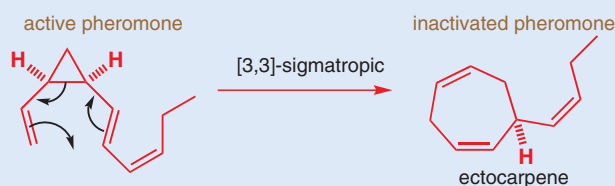
■ Notice that the product is a γ,δ -unsaturated carbonyl compound.

But the product of this rearrangement is now set up for a second [3,3]-sigmatropic rearrangement, this time made favourable by a shift into conjugation and the formation of two trisubstituted double bonds from two terminal ones. Overall, the prenyl group walks from one end of the molecule to the other, inverting twice as it goes.



Seaweed sex censored by a sigmatropic shift

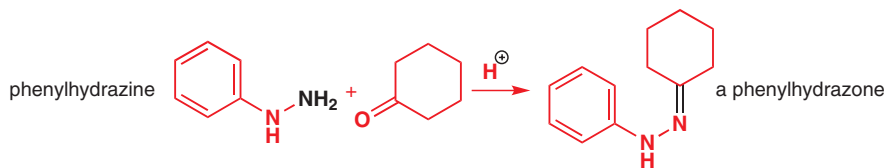
In order to reproduce, the female gametes of marine brown algae must attract mobile male gametes. This they do by releasing a pheromone, long thought to be the cycloheptadiene ectocarpene. In 1995 results were published that suggested that, in fact, the pheromone was a cyclopropane, and that ectocarpene was ineffective as a pheromone.



How had the confusion arisen? Well, the remarkable thing is that the cyclopropyl pheromone inactivates itself, with a half-life of several minutes at ambient temperature, by [3,3]-sigmatropic rearrangement to the cycloheptadiene, driven by release of strain from the three-membered ring. This not only confused the earlier pheromone chemists, but it also provides a marvellously precise way for the algae to signal their presence and readiness for reproduction without saturating the sea water with meaningless pheromone.

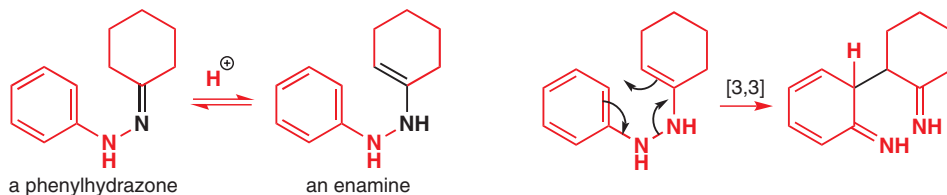
Applications of [3,3]-sigmatropic rearrangements using other elements

There is no need to restrict our discussion to carbon and oxygen atoms. We shall finish this section with two useful reactions that use other elements. You met the most famous synthesis of indoles in Chapter 30—the Fischer indole synthesis—and we can now look in more detail at the key step of this remarkable reaction. Condensation of phenylhydrazine with a ketone in slightly acidic solution gives a phenylhydrazone.



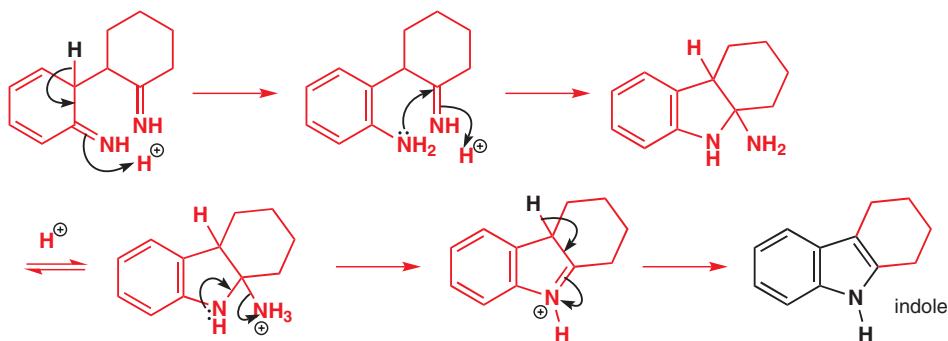
➡ Hydrazones—the imine derivatives of hydrazines—appeared in Chapter 11.

If the ketone is enolizable, this imine is in equilibrium with the corresponding enamine. The important bonds are given in black in the diagram. The enamine is ideally set up for a [3,3]-sigmatropic rearrangement in which the σ bond to be broken is the weak N–N σ bond and one of the π bonds is in the benzene ring.



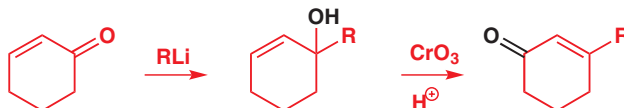
🖱 Interactive mechanism for Fischer indole synthesis

The product is a highly unstable double imine. But aromaticity is immediately restored and a series of proton shifts and C–N bond formation and cleavage reactions give the aromatic indole.

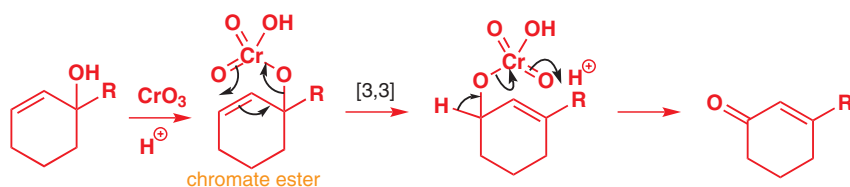


➡ A detailed discussion of this reaction as a synthesis of indoles appears in Chapter 30.

That was a [3,3]-sigmatropic reaction involving two nitrogens. There follows one with two oxygens and a chromium atom. When tertiary allylic alcohols are oxidized with CrO_3 in acid solution, no direct oxidation can take place, but a kind of conjugate oxidation occurs.



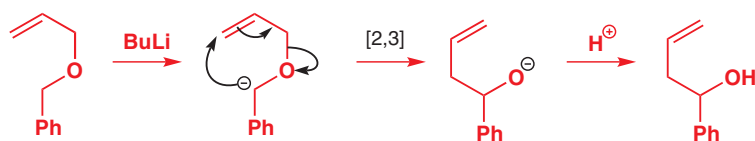
The first step in Cr(VI) oxidations can take place to give a chromate ester but this intermediate has no proton to lose so it transfers the chromate to the other end of the allylic system, where there is a proton. The chromate transfer can be drawn as a [3,3]-sigmatropic rearrangement. The final step is the normal oxidation in which chromium drops down from orange Cr(VI) to Cr(IV) and eventually by disproportionation to green Cr(III).



➔ Cr(VI) oxidations are described in Chapters 9 and 23.

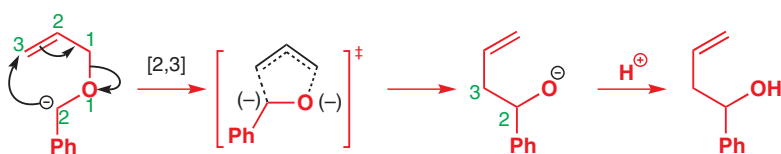
[2,3]-Sigmatropic rearrangements

All [3,3]-sigmatropic rearrangements have six-membered cyclic transition states. It is no accident that the size of the ring is given by the sum of the two numbers in the square brackets and this is universally the case for sigmatropic rearrangements. We are now going to look at [2,3]-sigmatropic rearrangements so we will be needing five-membered cyclic transition states. There is a problem here. You cannot draw three arrows going round a five-membered ring without stopping or starting on an atom, not a bond. This can be OK if the atom is a carbanion.

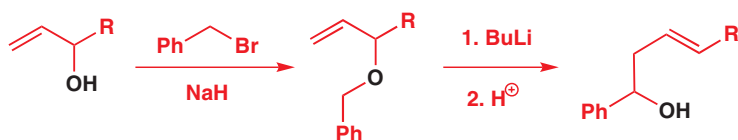


■ It can also be OK if the atom is a carbocation, or if it is an element that is happy to change oxidation state.

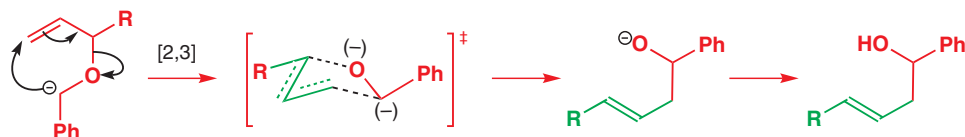
The starting material is a benzyl allyl ether and undergoes [2,3]-sigmatropic rearrangement to make a new C–C σ bond at the expense of a C–O σ bond—a bad bargain this as the C–O bond is stronger. The balance is tilted by the greater stability of the oxyanion in the product than of the carbanion in the starting material. The new bond has a 2,3 relationship to the old and the transition state is a five-membered ring.



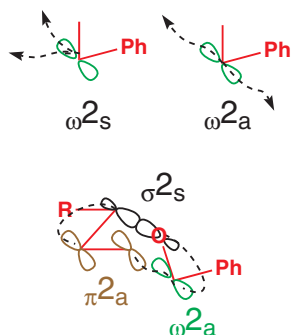
The transition state can be quite chair-like so that the new π bond will be *trans* if it has a choice. There will be a choice if the ether has been made from a substituted allyl alcohol.



We cannot draw a complete chair as we haven't got a six-membered ring, but the part that is to become the new π bond can be in a chair-like part of the five-membered ring. The substituent R prefers an equatorial position and the resulting *trans* arrangement of the groups is outlined in green.



We can use the same conformational diagram to show how the orbitals overlap as the new bond is formed. When we come to use the Woodward–Hoffmann rules on these [2,3]-sigmatropic rearrangements, we find something new. We have a π bond and a σ bond and a carbanion. How are we to represent a carbanion (or a carbocation) that is just a p orbital on an atom? The new symbol we use for a simple p orbital is ω (lower case omega).



➡ Much more sulfur chemistry is described in Chapter 27.

A carbanion is an ω_2 component and a carbocation is an ω_0 component as it has zero electrons. If the two new bonds are formed to the same lobe of the p orbital of the carbanion, we have an ω_2 component, but if they are formed to different lobes we have an ω_2 component.

Without going through the whole routine again, the [2,3]-sigmatropic rearrangement we have been discussing can be described as an $\omega_2 + \sigma_2 + \pi_2$ reaction. There is one $(4q+2)_s$ and no $(4r)_a$ component so the reaction is thermally allowed.

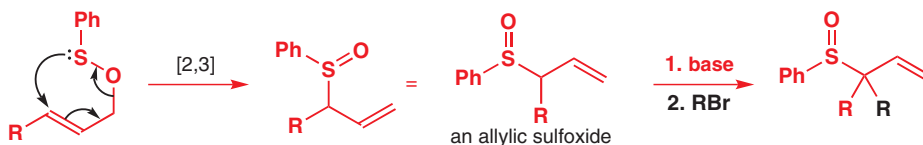
[2,3]-Sigmatropic rearrangements with S and Se

There are many [2,3]-sigmatropic rearrangements involving a variety of heteroatoms as well as carbon. The mechanism is common with elements that are prepared to change their oxidation state by two so that an arrow can both start and finish on that atom. The examples in this section involve sulfur and selenium, which can both form stable compounds at three oxidation states: S or Se(II), S or Se(IV), and S or Se(VI).

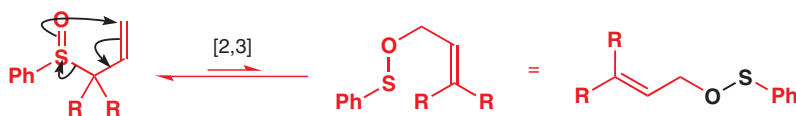


Reaction of an allylic alcohol with PhSCl gives an unstable sulfenyl sulfonate ester that rearranges on heating to an allylic sulfoxide by a [2,3]-sigmatropic rearrangement involving both O and S. Notice that arrows both start and stop on the sulfur atom, which changes from S(II) to S(IV) during the reaction. The new functional group with an S=O bond is a sulfoxide, and this is a good way of making allylic sulfoxides. The product forms an anion stabilized by sulfur, which can be alkylated.

Interactive mechanism for the [2,3]-sigmatropic shift of sulfoxides

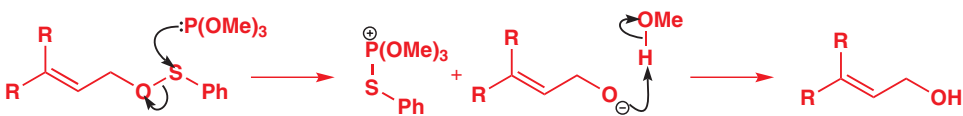
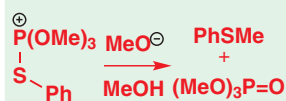


We have said that all these sigmatropic rearrangements are reversible but now we can prove it. If this product is heated in methanol with a nucleophile such as $(\text{MeO})_3\text{P}$ (trimethylphosphite), which has a liking for sulfur, the [2,3]-sigmatropic rearrangement runs backwards and a sulfonate ester is again formed.

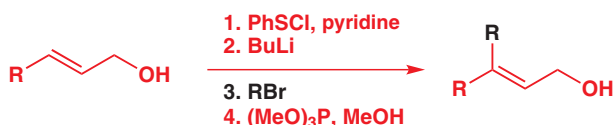


This is an unfavourable reaction because the equilibrium lies over on the sulfoxide side. But the nucleophile traps the sulfonate ester and the methanol ensures that the alkoxide ion formed is immediately protonated so that we get another allylic alcohol.

■ The other products are actually PhSMe and $(\text{MeO})_3\text{P}=\text{O}$. You might like to work out a mechanism for these stages of the reaction.

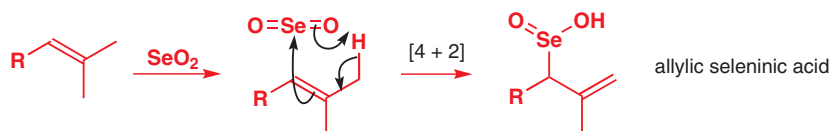


So what is the point of going round in circles like this? The net result is the alkylation of an allylic alcohol in a position where alkylation would not normally be considered possible.

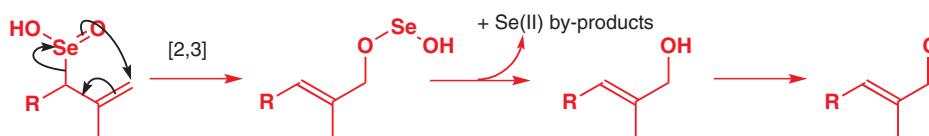


A related reaction of selenium in its +4 oxidation state (as selenium dioxide, SeO_2) allows us to make allylic alcohols and enals from simple alkenes. The overall reaction is the simple oxidation shown in the margin, but the route by which the compound gets there involves two successive pericyclic reactions.

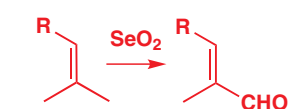
Selenium dioxide will react with alkenes in a [4 + 2] cycloaddition reminiscent of the ene reaction.



The initial product is an allylic seleninic acid—and just like an allylic sulfoxide (but more so because the C–Se bond is even weaker) it undergoes allylic rearrangement to give an unstable compound that rapidly decomposes to an allylic alcohol. In some cases, particularly this most useful oxidation of methyl groups, the oxidation continues to give an aldehyde or ketone.



Overall, CH_3 has been replaced by CH_2OH or $\text{CH}=\text{O}$ in an allylic position, a transformation similar to the allylic bromination reaction with NBS that you met in Chapter 24, but with a very different mechanism. The by-product of the oxidation is a selenium(II) compound, and it can be more practical to carry out the reaction with only a catalytic amount of SeO_2 , with a further oxidizing agent, *t*-butyl hydroperoxide, to reoxidize the Se(II) after each cycle of the reaction. This eliminates the need to get rid of large amounts of selenium-containing products, which are toxic and usually smelly.



➡ The ene reaction was introduced on p. 894.

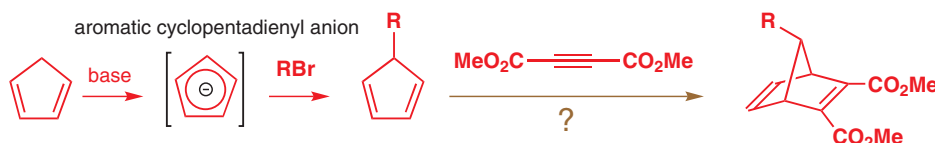
■ In a very few special cases, this seleninic acid intermediate has been isolated.

🔗 Interactive mechanism for allylic oxidation

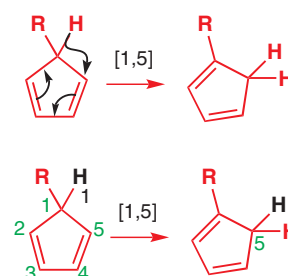
[1,5]-Sigmatropic hydrogen shifts

When one of the numbers in square brackets is '1', the old and new σ bonds are to the same atom, so we are dealing with the migration of a group around a conjugated system. In the case of a [1,5]-sigmatropic rearrangement the transition state is a six-membered ring (remember—just add together the numbers in square brackets). There is an important example in the margin. Let us first check that this is indeed a [1,5]-sigmatropic rearrangement by numbering the position of the new σ bond with respect to the old. Note that we must go the long way round the five-membered ring because that is the way the mechanism goes.

It is a [1,5]-sigmatropic rearrangement. The figure '1' in the square brackets shows that the same atom is at one end of the new σ bond as was at one end of the old σ bond. One atom has moved in a 1,5 manner and these are often called [1,5]-sigmatropic *shifts*. This is often abbreviated to [1,5]H shift to show which atom is moving. This particular example is important because sadly it prohibits a most attractive idea. The aromatic cyclopentadienyl anion is easily formed, stable, and readily alkylated. This sequence of alkylation and Diels–Alder reaction looks very good.

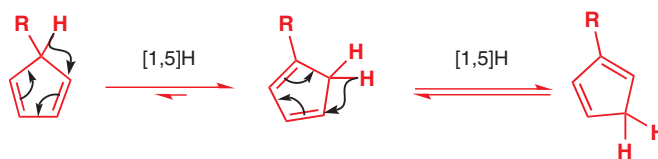


But sadly this sequence is, in fact, no good at all. A mixture of three Diels–Alder adducts is usually obtained resulting from addition to the three cyclopentadienes present in solution as

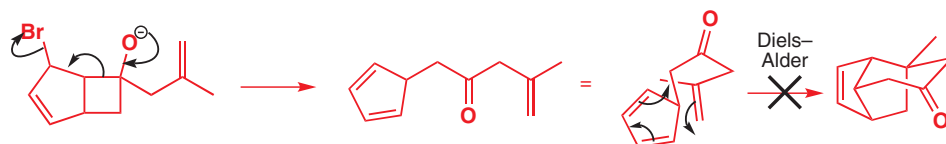


🔗 Interactive mechanism for a [1,5]-sigmatropic shift on cyclopentadiene

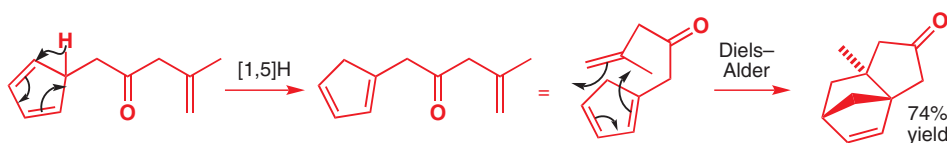
the result of rapid [1,5]H shifts. The one drawn above is a minor product because there is more of the other two dienes, which have an extra substituent on the double bonds.



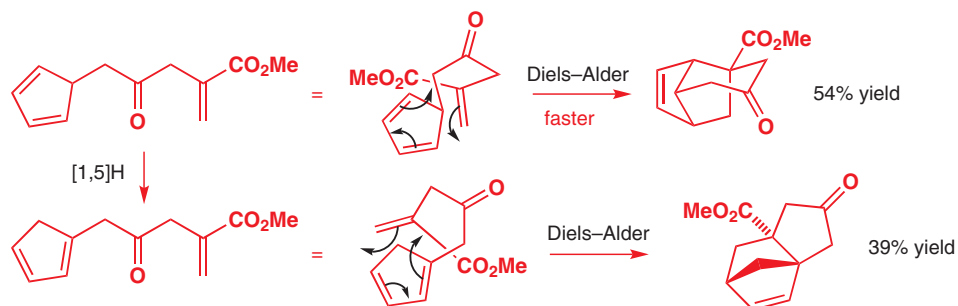
An excellent example comes from the intramolecular Diels–Alder reactions explored by Dreiding in 1983. One particular substituted cyclopentadiene was made by a fragmentation reaction (see Chapter 36). It might have been expected to give a simple Diels–Alder adduct.



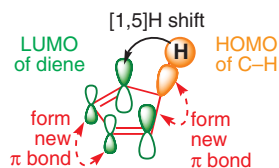
There is nothing wrong with this reaction—indeed, the product looks beautifully stable—but it is not formed because the [1,5]H shift is too quick and gives a more stable cyclopentadiene with more substituents on a double bond. *Then* it does the Diels–Alder reaction.



Notice that in these compounds the ketone is not conjugated to any of the alkenes and so does not influence the reaction. If we increase the reactivity of the dienophile by putting an ester group in conjugation with it, most of the compound does the Diels–Alder reaction *before* it does the [1,5]H shift.



Orbital description for the [1,5]H sigmatropic shift



■ You should satisfy yourself that the other frontier orbital combination—HOMO of the diene and LUMO of the C–H bond—works equally well.

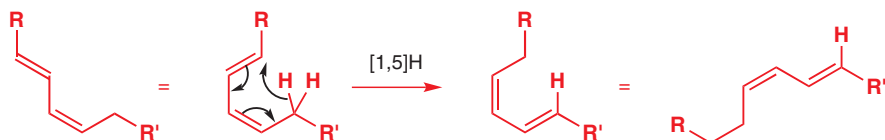
It is equally satisfactory to use frontier orbitals or the Woodward–Hoffmann rules for these reactions. We can take the diene as one component (HOMO or LUMO or π_4) and the C–H bond as the other (LUMO or HOMO or σ_2). Let us start by using the LUMO of the diene (ψ_3) and the HOMO of the C–H bond (its filled σ orbital), as shown in the margin. If the circle around the H atom surprised you, perhaps it will also remind you that hydrogen has only a 1s orbital, which is spherical. You can probably see already that all the orbitals are correctly lined up for the reaction.

The hydrogen atom slides across the top face of the planar cyclopentadiene ring. We call this a suprafacial migration, meaning that the migrating group leaves from one face of the π system and rejoins that same face (the top face in this example). Antarafacial migration would mean leaving the top face and rejoining the bottom face—a clear impossibility here.

If you use the Woodward–Hoffmann rules, you need to note that the hydrogen atom must react with retention. The 1s orbital is spherically symmetrical and has no node, so wherever you draw the dotted line from that orbital it always means retention. Choosing the components is easy—the diene is a π_4 and the C–H bond a σ_2 component.

The easiest way to join them up is to link the hydrogen atom's 1s orbital to the top lobe of the p orbital at the back of the diene and the black sp^3 orbital to the top lobe at the front of the diene. This gives us π_4 s and σ_2 s components and there is one $(4q+2)_s$ and no $(4r)_a$ components so the sum is odd and the reaction is allowed. Both approaches give us the same picture—a suprafacial migration of the hydrogen atom with (inevitably) retention at the migrating group.

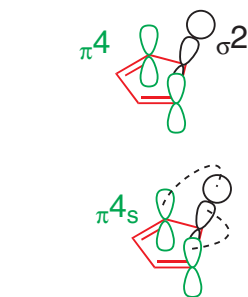
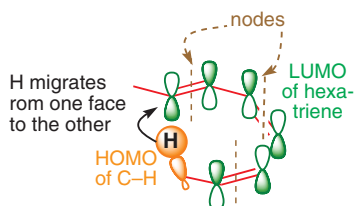
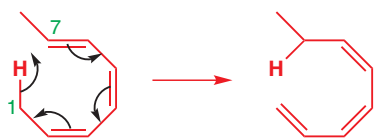
These [1,5]-sigmatropic shifts are not restricted to cyclopentadienes. In Chapter 34 we bemoaned the lack of Diels–Alder reactions using *E,Z* dienes. One reason for the shortage of examples is that such dienes undergo [1,5]H shifts rather easily and mixtures of products result.



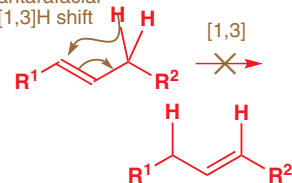
The consequences of orbital symmetry for sigmatropic hydrogen shifts are simple. In thermal reactions, [1,5]H shifts occur suprafacially but [1,3]H and [1,7]H shifts must be antarafacial. Antarafacial [1,3]H shifts are impossible, even though they are allowed, because a rigid three-carbon chain is too short to allow the H atom to transfer from the top to the bottom—the H atom just can't reach. This is just as well, as otherwise double bonds would just wander around molecules by repeated [1,3]H shifts.

When we come to [1,7]H shifts, the situation is different. Now the much longer chain is just flexible enough to allow antarafacial migration. The hydrogen atom leaves the top side of the triene and adds back in on the bottom side. The diagram shows this in orbital terms: the LUMO of hexatriene has three nodes. Antarafacial [1,7]H migration is allowed and possible.

allowed and possible antarafacial [1,7]H shift



allowed but impossible
antarafacial
[1,3]H shift



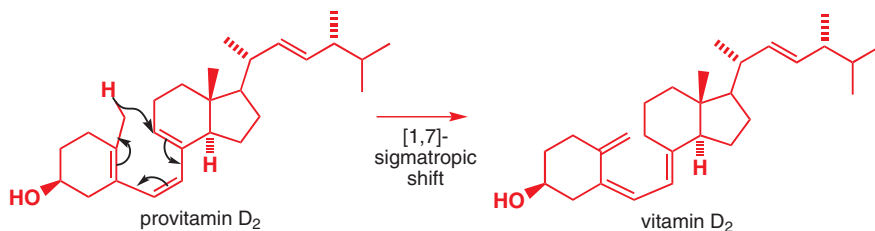
Summary of thermal sigmatropic migrations of hydrogen

	[1,3]H shift	[1,5]H shift	[1,7]H shift
stereochemistry	antarafacial	suprafacial	antarafacial
feasibility	impossible	easy	possible

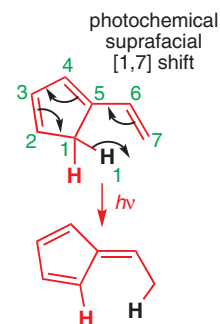
Photochemical [1,n] H sigmatropic shifts follow the opposite rules

As you should by now expect, all this is reversed in photochemical reactions. The margin shows an example of a [1,7]H shift that cannot occur antarafacially because the molecule is a rigid ring, but that can and does occur photochemically in a suprafacial manner.

A [1,7]H shift occurs in the final stages of the human body's synthesis of vitamin D from cholesterol. Here is the last step of the biosynthesis.

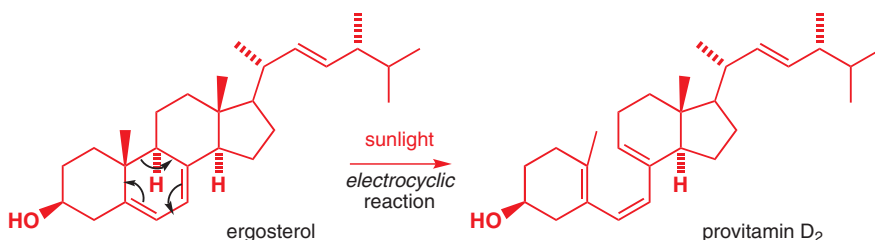


➡ The reversal of the rules of orbital symmetry when you move from thermal to photochemical reactions was described on p. 896.



This step happens spontaneously, without the need for light, so the [1,7]H shift must be antarafacial. That's no problem in this triene system—there is enough flexibility for the hydrogen atom to migrate from the top to the bottom face.

Why, then, does the body famously need sunlight to make vitamin D? The reason is the previous step, which can only occur when light shines on the skin.

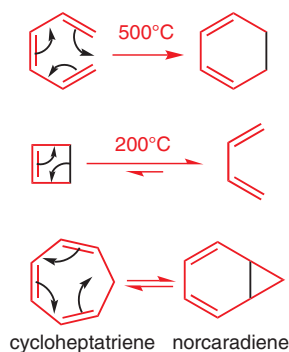
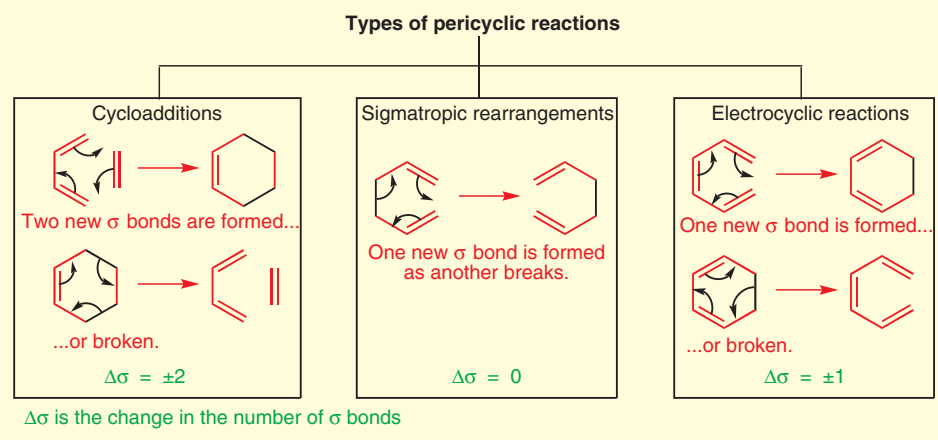


This ring opening is clearly pericyclic—the electrons go round in a ring, and the curly arrows could be drawn either way—but it is neither a cycloaddition (only one π system is involved) nor a sigmatropic rearrangement (a σ bond is broken rather than moved). It is, in fact, a member of the third and last kind of pericyclic reaction, an *electrocyclic reaction*.

Electrocyclic reactions

In an electrocyclic reaction a ring is always broken or formed. Rings may, of course, be formed by cycloadditions as well, but the difference with electrocyclic reactions is that just one new σ bond is formed (or broken) across the ends of a single conjugated π system. In a cycloaddition, two new σ bonds are always formed (or broken), and in a sigmatropic rearrangement one σ bond forms while one breaks.

- The types of pericyclic reactions are distinguished by the number of σ bonds made or broken



One of the simplest electrocyclic reactions occurs when hexatriene is heated to 500 °C. It is a pericyclic reaction because the electrons go round in a ring (you could equally draw the arrows going the other way); it's electrocyclic because a new σ bond is formed across the ends of a π system. The reaction goes because the σ bond that is formed is stronger than the π bond that is lost.

The opposite is true for the electrocyclic opening of cyclobutene—ring strain in the four-membered ring means that the reverse (ring-opening) reaction is preferred to ring closure.

In one famous case, the release of ring strain is almost exactly counterbalanced by the formation of a σ bond at the expense of a π bond. Cycloheptatriene exists in equilibrium with a bicyclic isomer known as norcaradiene. Usually cycloheptatriene is the major component of the equilibrium, but the norcaradiene structure is favoured with certain substitution patterns.

Rules for electrocyclic reactions

Whether they go in the direction of ring opening or ring closure, electrocyclic reactions are subject to the same rules as all other pericyclic reactions. With most of the pericyclic reactions you have seen so far, we have given you the choice of using either HOMO–LUMO reasoning or the Woodward–Hoffmann rules. With electrocyclic reactions, you really have to use the Woodward–Hoffmann rules because (at least for the ring closures) there is only one molecular orbital involved.

● Electrocyclic reactions

- An electrocyclic reaction is the formation of a new σ bond across the ends of a conjugated polyene or the reverse.

It is important that you do not confuse electrocyclic reactions with pericyclic reactions. Pericyclic is the name for the whole family of reactions involving no charged intermediates in which the electrons go round the outside of the ring. *Electrocyclic* reactions, *cycloadditions*, and *sigmatropic* rearrangements are the three main classes of *pericyclic* reactions.

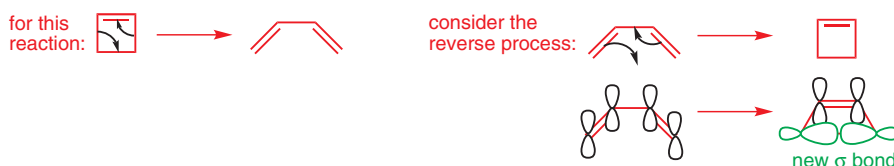
Let's start with the hexatriene ring closure from the beginning of this section, first looking at the orbitals and then following the same procedure that we taught you for cycloadditions and sigmatropic rearrangements to see what the Woodward–Hoffmann rules have to say about the reaction.

Hexatriene is, of course, a 6π electron ($\pi 6$) conjugated system and, on forming cyclohexadiene, the end two orbitals must rotate through 90° to form a σ bond.

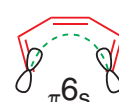
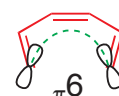
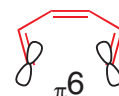
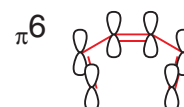
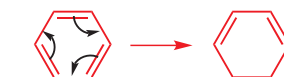
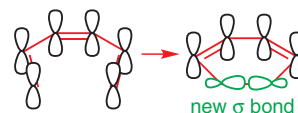
So, now for the Woodward–Hoffmann treatment.

1. Draw the mechanism for the reaction.
2. Choose the components. All the bonds taking part in the mechanism must be included and no others.
3. Make a three-dimensional drawing of the way the components come together for the reaction, putting in orbitals at the ends of the components (only!).
4. Join up the components where new bonds are to be formed. Make sure you join orbitals that are going to form new bonds.
5. Label each component s or a depending on whether new bonds are formed on the same or on opposite sides. We called this reaction 's' because the top halves of the two π orbitals join together.
6. Add up the number of $(4q+2)_s$ and $(4r)_a$ components. If the sum is odd, the reaction is allowed. Here there is one $(4q+2)_s$ component and no $(4r)_a$ components. Total = 1 so this is an allowed reaction.

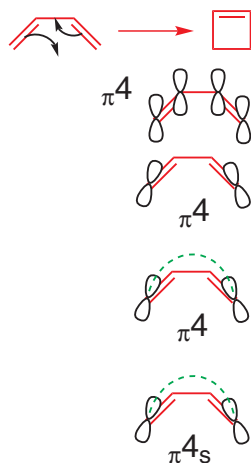
We can give the same treatment to the cyclobutene ring-opening reaction—the Woodward–Hoffmann rules tell us nothing about which way the reaction will go, only if the reaction is allowed, and it is usually easier with electrocyclic reactions to consider the ring-closing reaction even if ring opening is favoured thermodynamically. This is the process we need to consider:



■ In the same way, the Woodward–Hoffmann rules apply both to cycloadditions and to reverse cycloadditions, as you saw in Chapter 34.



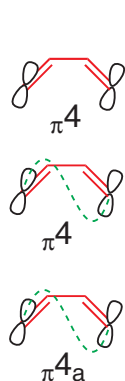
■ *Reminder.* In a thermal pericyclic reaction the total number of $(4q+2)_s$ and $(4r)_a$ components must be odd.



And the Woodward–Hoffmann treatment again.

1. Draw the mechanism for the reaction.
2. Choose the components. All the bonds taking part in the mechanism must be included and no others.
3. Make a three-dimensional drawing of the way the components come together for the reaction, putting in orbitals at the ends of the components (only!).
4. Join up the components where new bonds are to be formed. Make sure you join orbitals that are going to form new bonds.
5. Label each component s or a depending on whether new bonds are formed on the same or on opposite sides.
6. Add up the number of $(4q+2)_s$ and $(4r)_a$ components. If the sum is odd, the reaction is allowed. There are no $(4q+2)_s$ components and no $(4r)_a$ components. Total = 0 so this is a disallowed reaction.

Oh dear! We know that the reaction works, so something must be wrong. It certainly isn't Woodward and Hoffmann's Nobel-prize-winning rules—it's our way of drawing the orbital overlap that is at fault. We were fine up to stage 3 (we had no choice till then)—but see what happens if we make the orbitals overlap in a different way.

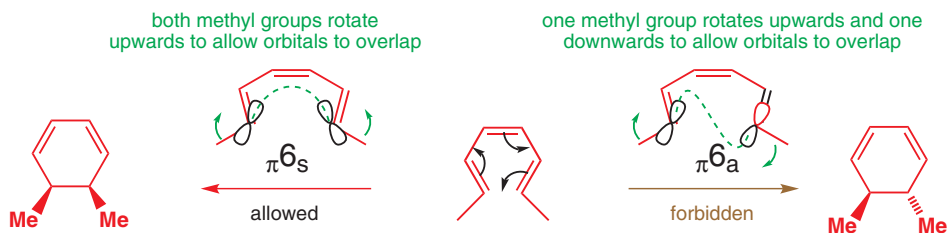


1. As before.
2. As before.
3. Make a three-dimensional drawing of the way in which the components come together for the reaction, putting in orbitals at the ends of the components (only!).
4. Join up the components where new bonds are to be formed. Make sure you join orbitals that are going to form new bonds.
5. Label each component s or a depending on whether new bonds are formed on the same or on opposite sides.
6. Add up the number of $(4q+2)_s$ and $(4r)_a$ components. If the sum is odd, the reaction is allowed. There are no $(4q+2)_s$ components and one $(4r)_a$ component. Total = 1 so this is an allowed reaction.

Now it works! In fact, extension of this reasoning to other electrocyclic reactions tells you that they are *all* allowed—provided you choose to make the conjugated system react with itself suprafacially for $(4n+2)\pi$ systems and antarafacially for $(4n)\pi$ systems. This may not seem particularly informative, since how you draw the dotted line has no effect on the reaction product in these cases. But it can make a difference. Here is the electrocyclic ring closure of an octatriene, showing the product from (a) suprafacial reaction and (b) antarafacial reaction.

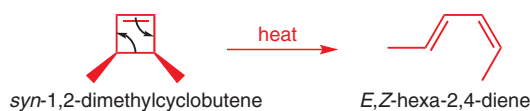
■ The green arrows in this and subsequent diagrams are merely mechanical devices to show the way in which the substituents move. They are nothing to do with real mechanistic curly arrows.

Interactive mechanism for disrotatory ring closure of hexatrienes

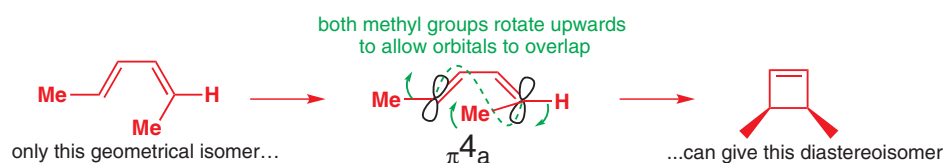


Conrotatory and disrotatory reactions

Whether the reaction is supra- or antarafacial ought to be reflected in the relative stereochemistry of the cyclized products—and indeed it is. The reaction above gives solely the diastereoisomer on the left, with the methyl groups *syn*—clear proof that the reaction is suprafacial. This is a difficult result to explain without the enlightenment provided by the Woodward–Hoffmann rules! This electrocyclic cyclobutene ring opening also gives the product as a single stereoisomer.



Again, if we draw the reverse reaction, we can see that the reaction required has to be antarafacial for the stereochemistry to be right.



We have drawn little green arrows on the diagrams to show how the methyl groups move as the new σ bonds form. For the allowed suprafacial reaction of the 6π electron system they rotate in opposite directions so the reaction is called **disrotatory** (yes, they both go up, but one has to rotate clockwise and one anticlockwise) while for the allowed antarafacial reaction of the 4π electron system they rotate in the same direction so the reaction is called **conrotatory** (both clockwise as drawn, but they might equally well have both gone anticlockwise). We can sum up the course of all electrocyclic reactions quite simply using these words.

 Interactive mechanism for conrotatory opening of cyclobutenes

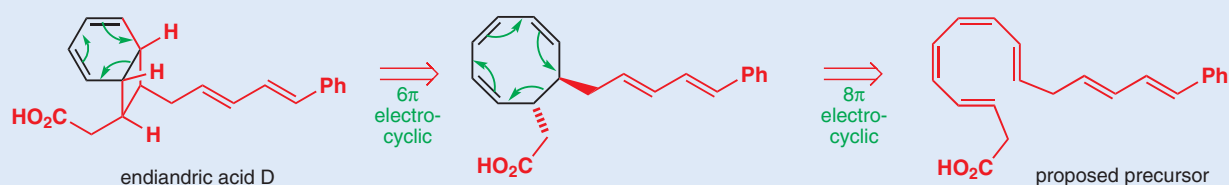
● Rules for electrocyclic reactions

- All electrocyclic reactions are allowed.
- Thermal electrocyclic reactions involving $(4n + 2)$ π electrons are *disrotatory*.
- Thermal electrocyclic reactions involving $(4n)$ π electrons are *conrotatory*.
- In *conrotatory* reactions the two groups rotate in the *same* turning sense: *both* clockwise or *both* anticlockwise.
- In *disrotatory* reactions, *one* group rotates *clockwise* and *one* anticlockwise.

This rotation is the reason why you must carefully distinguish electrocyclic reactions from all other pericyclic reactions. In cycloadditions and sigmatropic rearrangements there are small rotations as bond angles adjust from 109° to 120° and vice versa, but in electrocyclic reactions rotations of nearly 90° are required as a planar polyene becomes a ring or vice versa. These rules follow directly from application of the Woodward–Hoffmann rules—you can check this for yourself.

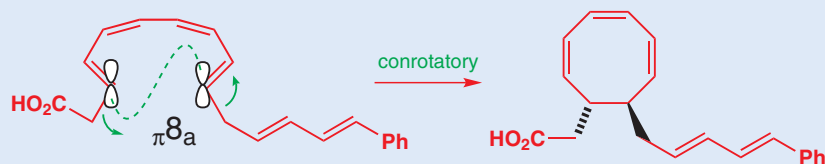
Electrocyclic reactions in nature: the endiandric acids

A beautiful example of electrocyclic reactions at work is provided by the chemistry of the endiandric acids. This family of natural products, of which endiandric acid D is one of the simplest, is remarkable in being racemic—most chiral natural products are enantiomerically pure (or at least enantiomerically enriched) because they are made by enantiomerically pure enzymes (we discuss all this in Chapter 42). So it seemed that the endiandric acids were formed by non-enzymatic cyclization reactions, and in the early 1980s their Australian discoverer, Black, proposed that their biosynthesis might involve a series of electrocyclic reactions, starting from an acyclic polyene precursor.



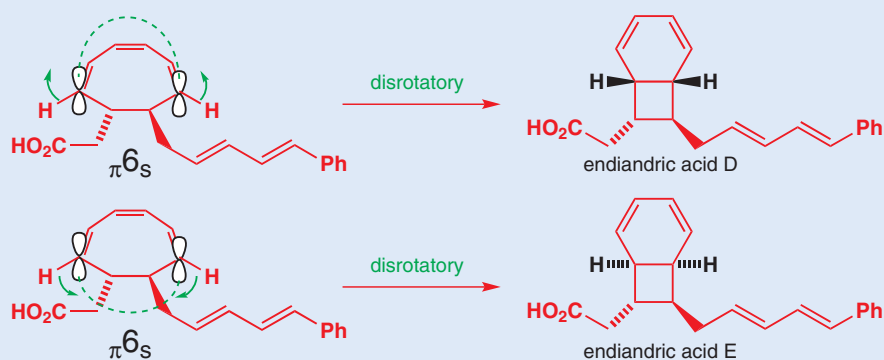
What made his proposal so convincing was that the stereochemistry of the endiandric acid D is just what you would expect from the requirements of the

Woodward–Hoffmann rules. The first step from the precursor is an 8π electrocyclic reaction, and would therefore be conrotatory.



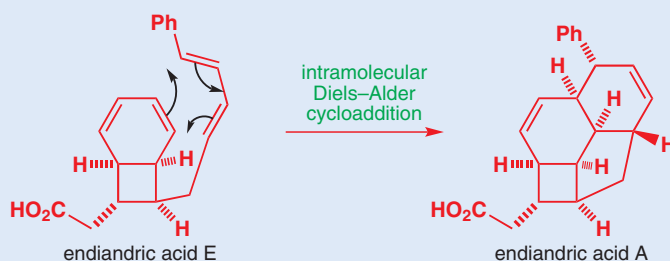
This sets up a new 6π system, which can undergo an electrocyclic reaction in disrotatory fashion. Because there are already chiral centres in the molecule, there are, in fact, two possible diastereoisomeric products from this reaction,

both arising from disrotatory cyclization. One is endiandric acid D; one is endiandric acid E.



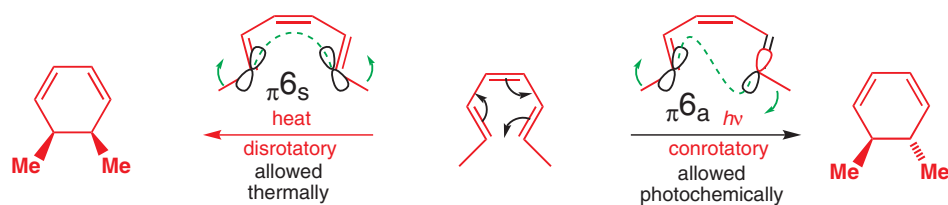
Of course, this was only a hypothesis—until in 1982 K.C. Nicolaou's group synthesized the proposed endiandric acid precursor polyene—and in one step made both endiandric acids D and E, plus endiandric acid A, which arises from a further pericyclic reaction—an intramolecular Diels–Alder cycloaddi-

tion of the acyclic diene on to the cyclohexadiene as dienophile. Endiandric acid A has four rings and eight stereogenic centres, and yet a series of pericyclic reactions produce it in one step from an acyclic polyene!

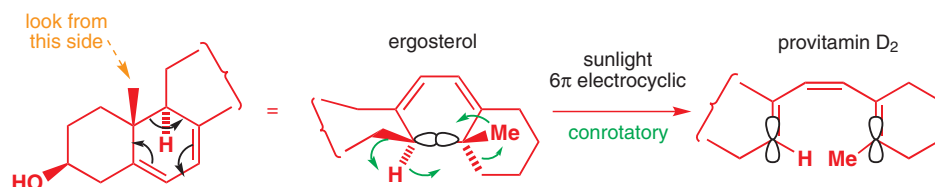


Photochemical electrocyclic reactions

After your experience with cycloadditions and sigmatropic rearrangements, you will not be surprised to learn that, in photochemical electrocyclic reactions, the rules regarding conrotatory and disrotatory cyclizations are reversed.



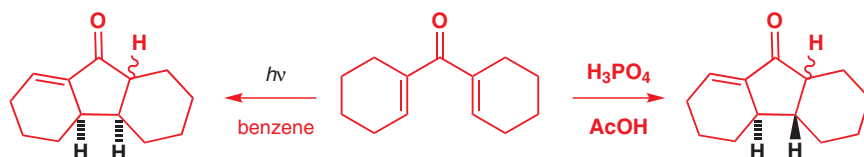
We can now go back to the reaction that introduced this section—the photochemical electrocyclic ring opening of ergosterol to give provitamin D_2 . By looking at the starting material and product we can deduce whether the reaction is conrotatory or disrotatory.



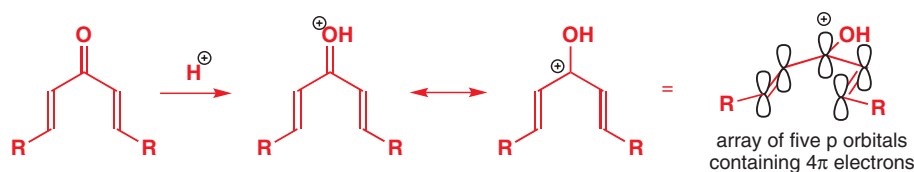
It's clearly conrotatory, and a little more thought will tell you why it has to be—a disrotatory thermal 6π cyclization would put an impossible *trans* double bond into one of the two six-membered rings. Vitamin D deficiency is endemic in those parts of the world where sunlight is scarce for many months of the year—and all because of orbital symmetry.

Cations and anions

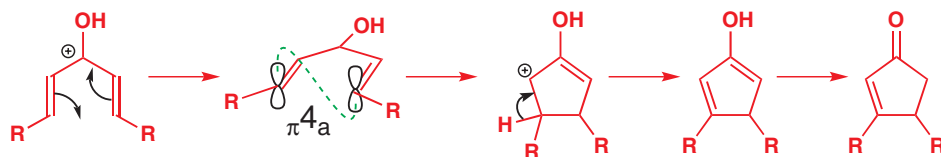
What we have just been telling you should convince you that the two reactions below are electrocyclic reactions, not least because the stereochemistry reverses on going from thermal to photochemical reaction.



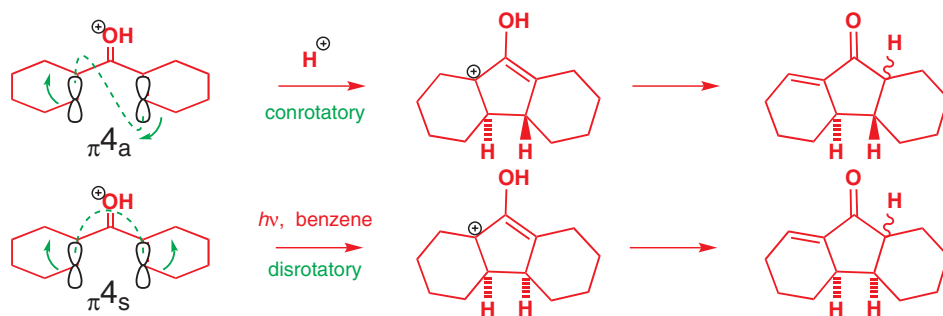
They are examples of what is known, after its Russian discoverer, as the Nazarov cyclization. In its simplest form, the Nazarov cyclization is the ring closure of a doubly α,β -unsaturated ketone to give a cyclopentenone. Nazarov cyclizations require acid, and protonation of the ketone sets up the conjugated π system required for an electrocyclic reaction.



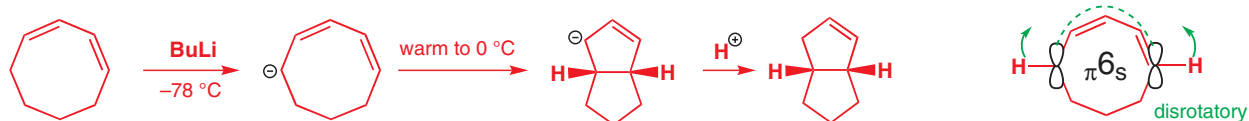
One of the five π orbitals involved is empty—so the cyclization is a 4π electrocyclic reaction, and the orbitals forming the new σ bond must interact antarafacially. Loss of a proton and tautomerism gives the cyclopentenone.



The example below confirms that the reaction is thermally conrotatory and photochemically disrotatory.



Dienyl cations and dienyl anions both undergo electrocyclic ring closure—a nice example occurs when this cyclooctadiene is deprotonated with butyllithium. There are still five p orbitals involved in the cyclization, but now there are six π electrons, so the reaction is disrotatory.



In this case, it is the conrotatory *photochemical* cyclization that is prevented by strain (it was tried—cyclooctadienyl anion is stable for at least a week at $-78\text{ }^{\circ}\text{C}$ in broad daylight) as the product would be a 5,5 *trans*-fused system. The same strain prevents thermal electrocyclic ring closure of cyclooctadienyl *cations*.

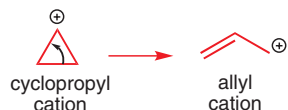
● **All electrocyclic reactions are allowed**

It would be a good point here to remind you that, although all electrocyclic reactions are allowed both thermally and photochemically providing the rotation is right, the steric requirements for con- or disrotatory cyclization or ring opening may make one or both modes impossible.

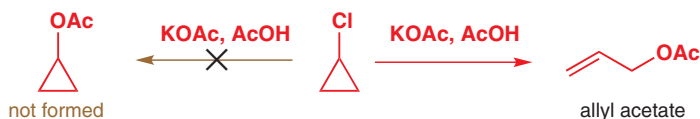
Small rings are opened by electrocyclic reactions

➡ You saw allyl cations as intermediates in substitution reactions in Chapters 15 and 24.

Ring strain is important in preventing a reaction that would otherwise change your view of a lot of the chemistry you know. Allyl cations are conjugated systems containing 2π electrons, so if you knew no other chemistry than what is in this chapter you might expect them to cyclize via disrotatory electrocyclic ring closure. The product would be a cyclopropyl cation.

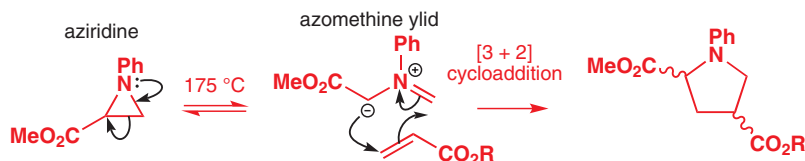


In fact, it is the cyclopropyl cations that undergo this reaction (very readily indeed—cyclopropyl cations are virtually unobservable) because ring strain encourages them to undergo electrocyclic ring opening to give allyl cations. The instability of cyclopropyl cations means that, even as they start to form as intermediates, they spring open to give allyl cation-derived products. Try nucleophilic substitution on a cyclopropane ring and this happens.

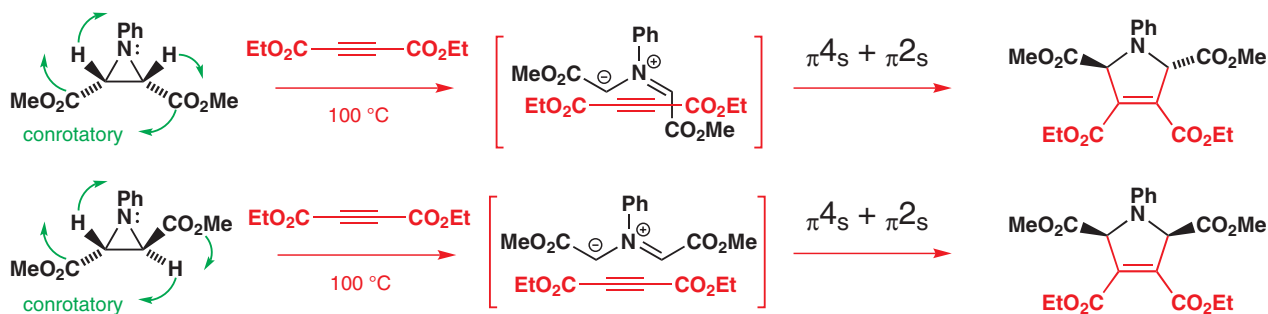


Electrocyclic ring opening of one type of three-membered ring tells us about the stereochemistry of the process. Many aziridines are stable compounds, but those bearing electron-withdrawing groups are unstable with respect to electrocyclic ring opening. The products are azomethine ylids and can be trapped by [3 + 2] cycloaddition reactions with dipolarophiles.

➔ You met 1,3-dipolar cycloadditions in Chapter 34, see p. 901.



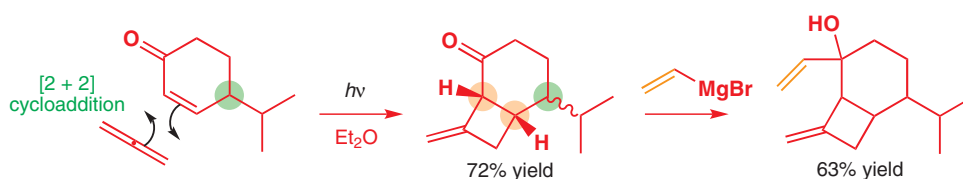
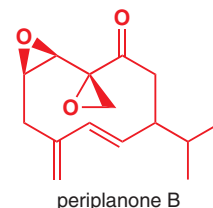
Because the cycloaddition is stereospecific (suprafacial on both components), the stereochemistry of the products can tell us the stereochemistry of the intermediate ylid, and confirms that the ring opening is conrotatory (the ylid is a 4π electron system).



The synthesis of a cockroach pheromone using pericyclic reactions

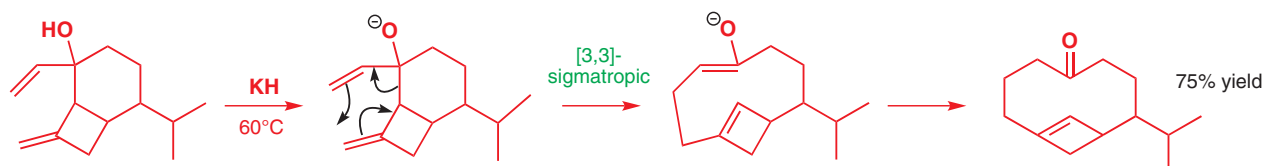
We finish this pair of chapters about pericyclic reactions with a synthesis whose simplicity is outclassed only by its elegance. Periplanone B is a remarkable bis-epoxide that functions as the sex pheromone of the American cockroach. Insect sex pheromones often have economic importance because they can form the key to remarkably effective traps for insect pests.

In 1984, Schreiber published a synthesis of the pheromone in which the majority of steps involve pericyclic reactions. Make sure you understand each one as it appears—re-read the appropriate part of Chapter 34 or this chapter if you have any problems. The first step is a photochemical [2 + 2] cycloaddition. You could not have predicted the regiochemistry, but it is typical of the cycloaddition of allenes with unsaturated ketones.



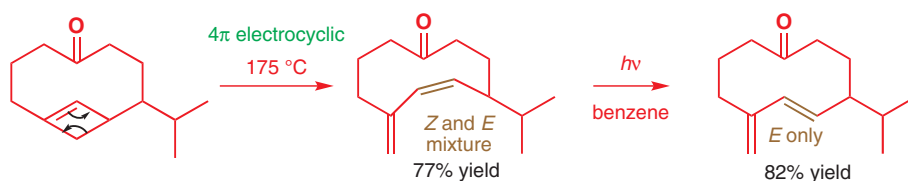
The product is a mixture of diastereoisomers because of the chiral centre already in the molecule (ringed in green), but it is, of course, fully stereospecific for the two new orange chiral centres in the four-membered ring. The next step adds vinylmagnesium bromide to the ketone—again a mixture of diastereoisomers results.

All the carbon atoms in the 12-membered ring are now present, and they are sorted out by the two steps that follow. The first is a Cope rearrangement: a [3,3]-sigmatropic rearrangement, accelerated as we have described (p. 914) by the presence of an alkoxide substituent.

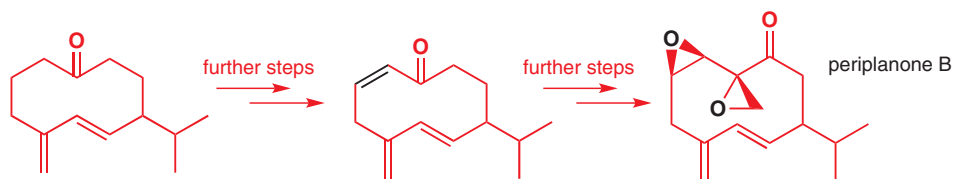


■ There are two things to note here—firstly the geometry of the double bond is nothing to do with whether the reaction is conrotatory or disrotatory. As you know this 4π electron electrocyclic ring opening must be conrotatory but as there is no substituent on the other end of the diene product we can't tell. Secondly notice that in this 12-membered ring a *trans* double bond is not only possible but probably preferred. We introduced irradiation as a means of interconverting double bond isomers in Chapter 27.

The six-membered ring has expanded to a ten-membered ring. Now for a second ring-expansion step—heating the compound to 175 °C makes it undergo electrocyclic ring opening of the four-membered ring, giving the 12-membered ring we want. Or rather not quite—the new double bond in the ring is formed as a mixture of *cis* and *trans* isomers, but irradiation isomerizes the less stable *cis* to the more stable *trans* double bond.



The remaining steps in the synthesis involve the insertion of another *Z* alkene and two epoxides. Pericyclic reactions are particularly valuable in the synthesis and manipulation of rings.



We must now take our leave of this trio of pericyclic reactions and move on to two reaction classes that have appeared frequently in these two chapters, but that also involve mechanisms other than pericyclic ones and deserve a chapter of their own: *rearrangements* and *fragmentations*.

Further reading

For explanations of pericyclic reactions and other reactions, using the full molecular orbital treatment, consult: Ian Fleming, *Molecular Orbitals and Organic Chemical Reactions, Student Edition*, Wiley, Chichester 2009. There is also a more comprehensive edition intended for practicing chemists, called the *Library Edition*. He has also written an Oxford Primer: *Pericyclic Reactions*, OUP, Oxford, 1999.

For a comprehensive treatment of sigmatropic and electrocyclic reactions in the synthesis of nitrogen heterocycles, see P. Wyatt and S. Warren, *Organic Synthesis: Strategy and Control*, Wiley, Chichester, 2007, chapter 34.

The synthesis of periplanone appears in S. L. Schreiber and C. Santini, *J. Am. Chem. Soc.*, 1984, **106**, 4038.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>